

analogue gyroscope that accurately measures angular rates along three orthogonal axes. The 3-axis MEMS (micro-electro-mechanical systems) gyroscope heralds the widespread adoption of realistic-motion user interfaces in all kinds of consumer applications.

The sensor provides 360° angular-rate detection for high-precision 3D gesture and motion recognition in mobile phones, game controllers, personal navigation systems and other portable devices. Combined with a 3-axis accelerometer, the gyroscope enables the creation of inertial measurement units, devices that track and deliver complete information on the type, rate and direction of motion of humans, vehicles and other objects.

First optical frequency comb

Scientists at University of Konstanz in Germany and National Institute of Standards and Technology (NIST) in the United States have built the first optical frequency comb — a tool for precisely measuring different frequencies of visible light — that actually looks like a comb.

The 'teeth' of the new frequency comb are separated enough that when viewed with a simple optical system — a grating and microscope — the human eye can see each of the approximately 50,000 teeth spanning the visible colour spectrum from red to blue.

A frequency comb with such well-separated, visibly distinct teeth will be an important tool for a wide range of applications in astronomy, communications and many other areas.

SixthSense going open source!

Pranav Mistry, a 28-year-old research assistant at Massachusetts Institute of Technology's Media Lab, has demonstrated the latest version of 'SixthSense' — a wearable gestural interface that augments the physical world with digital information and let us use

Google to introduce unified communication platform

Google will soon introduce a 'unified' communication platform named Google Wave that glues together a number of things—including e-mail, instant messaging, blogging, document sharing, wikis and multimedia content—which have been until now separate products.

"In Google Wave, you can create a Wave and add people to it. Everyone on your Wave can use formatted text, photos, gadgets and even feeds from other sources on the Internet. They can insert a reply or edit the wave directly. It is concurrent rich-text editing, where you can see instantly what your fellow collaborators are typing in your Wave," says Lars Rasmussen, software engineering manager, Google.

Wave has been constructed on Google Web Toolkit using HTML 5 and has a rich set of open APIs that allow developers to embed waves in other Internet services and to build extensions that work inside them. This means developers can use and enhance Google Wave through two types of developments: By building robot extensions to automate common tasks or building gadget extensions to provide a new way for users to interact.

Google Wave is still in active development and is expected to remain in development until later in 2009.

—Monika Bhati



Pranav Mistry with SixthSense prototype

computing device in the user's pocket. The system projects information onto the surfaces and physical objects around us, making any surface into a digital interface. The camera recognises and tracks both the user's hand gestures and physical objects using computer-vision-based techniques.

Internet gets a desi flavour!

Now, users can type Web addresses in Hindi, Tamil, Urdu, Bengali, Gujarati, Punjabi and Telegu. This has become possible with the Internet Corporation for Assigned Names and Numbers (ICANN), which allows the use of seven Indian languages in Web addresses.

At present, Internet addresses are limited to Latin characters from 'A to Z.' Domain names—.com, .net, .org, .in—are used as identification labels.

More languages would be added in the near future. The Department of Information Technology has started giving free fonts in 22 official Indian languages.

natural hand gestures to interact with that information. Mistry now plans to open-source his project (make the code freely available).

The SixthSense prototype comprises a pocket projector, mirror and camera worn in a pendant-like mobile device. Both the projector and the camera are connected to a mobile

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